DRILL 23

VEHICLE EXTRICATION

NFPA 1001 Level 2 (JPR 5.4.1) / Rural Rescue

HYDRAULIC RESCUE TOOLS – DOOR POP & HINGE REMOVAL

⏱ **Time:** 60–75 min | 🚒 **Apparatus:** Rescue / Engine | 🦺 **PPE:** Extrication Gloves, Bunker Gear, Helmet & Safety Glasses

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Tool Operator, Backup Firefighter & Patient Shield

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to safely operate hydraulic extrication tools (power unit, spreaders, and cutters—hydraulic hose or battery e-draulic), create a purchase point on a jammed vehicle door, force the door latch or shear the hinges, and remove the door completely to create a wide patient extrication path within **5 minutes**.

📦 TOOLS & APPARATUS SETUP

- Hydraulic rescue tool set: Spreader, cutter, hydraulic power unit (if hydraulic) or fresh batteries (if battery e-draulic).
- Scrap training vehicle with intact doors.
- Halligan bar (for creating initial gap if needed).
- Patient shield (hard polycarbonate board or heavy turnout blanket).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Tool Reaction & Pinch Zone:** Hydraulic spreaders exert up to 40,000 lbs of force. As the tips spread, the tool body can kick upward or pivot violently into the operator's thighs or ribs. Operators must position their bodies **outside the tool's swing arc**.

⚠️ **Hand Guards:** Keep gloved hands strictly behind the safety hand-guard. Never place fingers near the moving arms or cutting blades!

⚠️ **Pressurized Hose Line:** When operating hydraulic or fuel-powered rescue tools around motor vehicles, a **charged 1.75" dry chemical or foam line** must be staffed and staged as a fire suppression standby.

🛑 **THE STOP RULE: Shout "STOP!" immediately if tool tips slip out of a purchase point, or if the tool body begins to pinch the operator against the car frame.**

PHASE 1: TOOL INSPECTION & POWER UNIT SETUP (10 MINUTES) — SLOW DRY RUN

1. **Hydraulic Hose Connection (Core Coaxial / Dual Hose):**
   - Inspect couplings for grit/metal shavings. Connect and twist locking collar.
   - Start hydraulic gas power unit; verify choke off, engine at operating throttle, and control valve set to "Pressure".
2. **Battery (e-Draulic) Verification:**
   - Insert charged lithium-ion battery until latch clicks; check battery indicator (must show 4 green LEDs).
3. **Control Handle Familiarization:**
   - Modern tools use a deadman twist throttle: rotate right to open/spread, rotate left to close/cut. Releasing the handle automatically locks arms in place.

DRILL 23: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Vehicle Extrication • Practical Application

PHASE 2: CREATING THE PURCHASE POINT (15 MINUTES) — BUDDY COACHING

1. **Method 1 (Pinching the Window Frame / Rocker):**
   - Place spreader tips on the metal door skin near the top latch or bottom corner.
   - Squeeze/pinch the sheet metal inward to buckle the door edge away from the N-pillar, creating a 1-to-2-inch gap.
2. **Method 2 (Mechanical Purchase with Halligan):**
   - Drive the adz of a Halligan bar between the door edge and B-pillar at a 45-degree angle to pry the seam open.

PHASE 3: THE DOOR POP (LATCH SIDE VS HINGE SIDE) (25 MINUTES) — TIMED / STRESS

1. **Popping the Latch (B-Pillar Side):**
   - Insert spreader tips fully into the purchase gap (full tip insertion prevents tips from spitting out!).
   - Rotate handle to spread. Watch the sheet metal tear.
   - Work the tips downward toward the heavy **Nader pin (latch)** mechanism.
   - Spread with the full width of the arms. The steel latch mechanism will snap with a loud "BANG!"
2. **Shearing or Spreading the Hinges (A-Pillar Side):**
   - Once the latch is popped, swing the door wide open.
   - To remove the door completely:
   - **Option A (Cutters):** Place hydraulic cutter blades perpendicular to the door hinges. Cut the upper hinge completely, then cut the lower hinge and wire harness.
   - **Option B (Spreaders):** Insert spreader tips directly into the hinge pocket against the car frame and spread; the hinge bolts will shear cleanly off the chassis.
3. Remove the door completely from the vehicle and carry it to the designated scrap pile (sharp edges taped or covered).

⚠ COMMON PITFALLS & FIREGROUND HAZARDS

- ✗ *Spreading with only the tip ends:* Using only the very tips of the spreader arms causes the tool to spit out under load with explosive force. Always bury the tips deep into the seam!
- ✗ *Cutting high-strength boron steel at an angle:* Twisting the cutter blades while shearing a modern ultra-high-strength steel B-pillar will crack or shatter the tool blades. Always cut **at a clean 90-degree angle** to the post!
- ✗ *Standing between the tool and the car:* If the spreader tips slip, the 50-lb tool body will kick directly into your chest or groin. Stand to the side!

🗣 5-MINUTE TAILGATE DEBRIEF

1. ***"Did the spreader tips stay seated without slipping?", "How did the operator position their body relative to the tool kick zone?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Hydraulic power unit shut down; pressure dumped using dump valve before uncoupling hoses.
- ☐ Cutter blades inspected for nicks, cracks, or blade spread.
- ☐ Attendance roster signed.
- ☐ Spreader tips left slightly cracked (1/4 inch open, not touching) to relieve internal hydraulic pressure during thermal changes.
- ☐ Tools wiped clean of motor oil and broken glass.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS